CptS 484: Software Requirements

WRS Evolution

Phase I: Requirements Elicitation (Initial Understanding)

Project THEIA

Indoor Navigation Assistant for Blind and Visually Impaired Users

Team EV · Fall 2026 · Version 1.1 · October 18, 2026

Team member	Role in Phase I
Zichuo Wang	Team liaison; Project Plan owner; WRS §3 co-author; AI Usage and Teamwork Report owner; prototype co-author; document assembly
James Zuniga	AS-IS/TO-BE presentation owner; WRS §2 co-author
Austin Aragon	WRS §2 owner; AS-IS/TO-BE presentation co-author
Young Kim	Prototype mock-up and User Manual owner; Project Plan co-author; AI report co-author
Justus Jones	WRS §3 owner; independent §3 draft (Oct 18), from which the priorities, acceptance criteria, error handling, and Preliminary Definition coverage table (§5.4) were merged

Team repository: https://github.com/nemonto/cpts484-theia-ev

Presentation video: https://youtu.be/0aHIRkDOK3E

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B. Revision History

Date	Version	Changes	Editor(s)
2026-09-21 – 09-27	0.1	Candidate issues collected from all members in #wrs (each member at least 3)	All contributing members
2026-10-13	0.2	Issues list consolidated into the §2 draft (`Issues.md`)	Austin Aragon
2026-10-14	0.3	Front matter, §2 options/decisions, and §3 Improved Understanding drafted as templates	Zichuo Wang (AI-assisted)
2026-10-15	0.4	Prototype mock-up and preliminary User Manual drafted (§4)	Young Kim, Zichuo Wang
2026-10-16	0.5	§2 finalized with options, decisions, and rationale	Austin Aragon, James Zuniga
2026-10-17	0.6	§3 finalized and reconciled with the AS-IS/TO-BE slides, prototype, and manual; §4, §5 completed; document assembled	Zichuo Wang, Young Kim
2026-10-18	1.0	Final Phase I proofing and submission	Zichuo Wang, all review
2026-10-18	1.1	Merged additions from Justus Jones's independent §3 draft: priorities for all requirements, acceptance criteria for NFR1–NFR7, FS1.3–FS1.4, FS3.6, FS8.2–FS8.3, FS10.7, and §5.4 Preliminary Definition coverage. Roles updated (§C.1)	Zichuo Wang (from Justus Jones's draft)

C. Process

C.1 Team organization

Team EV has five members. Each work product has one **owner**, who is responsible for it, and one **co-author**, who writes part of it and can take it over if needed. Every member is expected to review every work product. The full assignment is in the Project Plan, §2.3.

Work product	Owner	Co-author
AS-IS/TO-BE presentation and video	James Zuniga	Austin Aragon
WRS §2 Issues	Austin Aragon	James Zuniga
WRS §3 Improved Understanding and traceability	Justus Jones	Zichuo Wang
Prototype mock-up and User Manual	Young Kim	Zichuo Wang
Project Plan	Zichuo Wang	Young Kim
AI Usage and Teamwork Report	Zichuo Wang	Young Kim

Changes from the preliminary plan. James Zuniga replaced Justus Jones as the WRS §2 co-author, Zichuo Wang replaced Young Kim as the WRS §3 co-author, and Young Kim replaced James Zuniga as the Project Plan co-author. Justus Jones, the WRS §3 owner, did not reply to the team during the final week, so Zichuo Wang completed §3 for the document assembled on Oct 17. Justus then wrote an independent §3 draft from WRS §2 (WRS-Section3-Updated.md, Oct 18). The team checked it against the slides, the prototype, and the manual, and merged its additions into this version: a priority for every requirement, acceptance criteria for the non-functional requirements, error handling in FR1, FR3, FR8,

and FR10, and the Preliminary Definition coverage table (§5.4). Parts that conflicted with the recorded scenarios or the prototype were not adopted.

C.2 Meetings and communication

The course is online and every member has a full course load, so the team worked **asynchronously**. There were no live meetings; instead:

- **Discord** was the main channel, with a channel per work product (e.g., #wrs). Issues, scenario ideas, drafts, and review feedback were posted there.
- **Weekly written checkpoints** (Sundays) were posted by the liaison. Each one summarized progress, assigned the next week's tasks, and asked members to review.
- **GitHub repository** holds the working drafts and the submitted versions.

C.3 Inputs

- Cpt S 484 Team Project Specification, Phase I, v2.0
- Cpt S 484 Project THEIA Preliminary Definition, v1.0
- World Assumptions Masterlist, v1.0
- CPTS 484 FAQ
- WRS template, WRS sample documents, and the Requirements Review Checklist
- AS-IS/TO-BE presentation template (Theia-AS-IS_and_TO-BE_v2.0.pptx)

C.4 Activities

The team followed the spiral requirements engineering process (Kotonya & Sommerville):

Step	Activity	Output
Elicitation	Members read the Preliminary Definition and the Masterlist and posted at least 3 candidate issues and a scenario idea in Discord. No real blind users were available to interview, so members reasoned from the position of a potential user.	Raw issues list; scenario ideas
Analysis and negotiation	Duplicate issues were merged and sorted into §2.1–2.3. For each issue the team listed options and chose one with a rationale. Out-of-scope items were decided against the Masterlist.	WRS §2
Documentation	Problems and Goals were derived from the issues, broken down into functional and non-functional objectives, then into requirements and specifications with traceability. The same decisions were used to write four AS-IS/TO-BE scenario pairs, the mock-up, and the user manual.	WRS §3, slides, §4
Validation	Drafts were reviewed against the Requirements Review Checklist and checked for consistency across the WRS, the slides, the prototype, and the manual before submission. Where they disagreed, the recorded slides were treated as fixed and the WRS was aligned to them; the prototype and manual were then aligned to the WRS.	Final Phase I documents

C.5 Resources and tools

Discord, Google Drive, GitHub, Microsoft PowerPoint/Word, and generative AI tools (ChatGPT, Gemini / NotebookLM, Claude / Claude Code). AI was used for drafting, critique, and assembly, and all AI output was reviewed by team members; details and evidence are in the AI Usage and Teamwork Report.

1. Introduction

1.1 Purpose

Blind and visually impaired people must often find their way through unfamiliar or changing indoor spaces: from a classroom to an office, a lab, or a restroom, sometimes on another floor or in a connected building. A white cane detects nearby obstacles and a guide dog helps avoid them, but neither tells the user where they are, which way to go, when to turn, or how to get help with an exact location when they are lost.

THEIA is a smartphone app that helps blind and visually impaired users navigate indoors using voice, gestures, audio, and vibration. This document records Team EV's requirements engineering work for Phase I. It starts from the informal THEIA Preliminary Definition, identifies its issues and resolves them (§2), and builds an improved, traceable model of the World, Requirements, and Specifications (§3). It also includes a preliminary prototype and user manual (§4). It is intended for the course instructor and the team, and it is the basis for the Phase II prototype.

1.2 Scope

In scope. THEIA supports a blind or visually impaired user (student, staff, or visitor) moving between rooms and significant locations inside mapped university campus buildings. This includes moving between floors and through mapped indoor connectors between buildings. Main functions:

- telling the user their current location,
- accepting a destination by voice and offering route options,
- step-by-step walk, turn, and stop guidance,
- warnings about known static obstacles and rerouting around blockages,
- guidance to elevators and stairs, with floor confirmation,
- off-route detection and recovery,
- help and emergency contact that sends the user's location,
- personal settings and navigation modes (Fast, Safe, Comfortable),
- suggestions based on the user's schedule.

Caretakers configure safety settings and receive alerts. Campus accessibility staff maintain the building maps.

Out of scope (based on the World Assumptions Masterlist):

- outdoor navigation between unconnected buildings,
- navigation inside a room (finding a chair or desk),
- detecting moving obstacles such as pedestrians,
- operating elevators or stairs for the user.

THEIA assumes beacons at each room and significant location, and it complements a cane or guide dog rather than replacing them. No working software is delivered in Phase I; the prototype is a mock-up.

1.3 Objectives and Success Criteria

THEIA should let a blind user:

1. know where they are at any time inside a mapped building;
2. reach a chosen destination, including on another floor or in a connected building, without sighted help;
3. receive clear and timely walk, turn, and stop instructions, and warnings about known obstacles;
4. recover when off route, and get help with an exact location when needed, without false alarms;
5. adjust the app to their own needs by voice, while caretakers control the safety settings;
6. use every function without seeing the screen.

This Phase I document is successful if:

- every issue in §2 has options, a decision, and a rationale;
- every Goal in §3 traces back to a Problem and forward to an Objective, and every requirement traces to an Objective;
- the four AS-IS/TO-BE scenarios, the prototype, and the user manual are consistent with the requirements in §3.

1.4 Definitions, Acronyms, and Abbreviations

Term	Definition
THEIA	The indoor navigation app being specified (named after the Greek goddess of vision).
Primary user	A blind or visually impaired person using THEIA to navigate.
Caretaker	A family member or assistant who configures safety settings and responds to alerts.
Beacon	A small wireless (Bluetooth) transmitter at a room or significant location that broadcasts its identity to the phone.
Building map	Digital floor plans with beacon positions, walkable segments, static obstacles, and closures, maintained by accessibility staff.
Connector	An indoor passage joining two buildings (skybridge, tunnel, shared hallway).
Off route	The user is no longer on the planned route: an unexpected beacon is detected, or the expected one is not detected in time.
Static obstacle	A non-moving object on a route (pillar, bench, trash can, construction barrier).
Haptic	Feedback by vibration.
Navigation mode	Fast, Safe, or Comfortable route preference selected by the user.
WRS	World, Requirements, Specification: the model relating the domain (W) to the system's requirements (R) and specifications (S).
FR / NFR	Functional Requirement / Non-Functional Requirement.
FS	Functional Specification (detailed, verifiable behavior under an FR).
Priority	How much a requirement matters: High is needed for a user to complete a trip safely, Medium improves it, Low can wait for Phase II.
DI / FRI / NFRI	Domain Issue / Functional Requirement Issue / Non-Functional Requirement Issue (§2).
P / G	Problem / Goal (§3.1).
IFO / INFO	Improved Functional Objective / Improved Non-Functional Objective (§3.1.3).
ID / WA	Improved Domain description / World Assumption (§3.1.3.1).
AS-IS / TO-BE	A scenario describing how a task is done today / how it would be done with THEIA.

Term	Definition
S1–S4	The four AS-IS/TO-BE scenario pairs: S1 Basic step-by-step guidance (Stevie), S2 Off-route recovery and emergency escalation (Alfred, Top Priority), S3 Multi-floor navigation by elevator (Mark), S4 Personal settings and navigation modes (Elena).

1.5 Overview

- **§2 Issues with the Preliminary Definition Given:** issues found in the Preliminary Definition (ambiguity, incompleteness, inconsistency, redundancy, unsoundness), each with options, a decision, and a rationale. §2.1 covers the domain, stakeholders, and objectives; §2.2 the functional requirements; §2.3 the non-functional requirements.
- **§3 WRS:** §3.1 (World) states the Problems and Goals and the improved understanding of the domain, world assumptions, stakeholders, and functional and non-functional objectives. §3.2 (RS) gives the functional and non-functional requirements and their specifications, each traced to the issues and objectives it satisfies.
- **§4 Preliminary Prototype and User Manual:** the mock-up of THEIA's voice, gesture, and vibration interface, and a user manual organized around the TO-BE scenarios.
- **§5 Traceability:** summary matrices linking Problems → Goals → Objectives → Requirements → Issues → Scenarios → Prototype screens. §5.4 maps every statement of the Preliminary Definition to the WRS.

The References list the course materials and the team's other Phase I work products.

2. Issues with the Preliminary Definition Given

2.1 Issues with "Domain, Stakeholders, Functional and Non-Functional Objectives"

2.1.1 DI1: Stakeholder roles

Original statement	"Staff members in the accessibility department and the police might also be considered secondary stakeholders."
Classification	Ambiguous, Incomplete
Issue	Accessibility staff and police are listed as stakeholders, but nothing says what they do in the system or what the system expects from them. Nothing says who creates or maintains the building maps either, though navigation can't work without them.
Option 1	List them as stakeholders only, with no defined role.
Option 2	Give each stakeholder a concrete responsibility tied to an app function.
Decision	Option 2
Rationale	Every stakeholder should justify a requirement. Accessibility staff manage spatial data (maps, temporary closures); Facilities manage physical infrastructure (beacons); Campus Police receive emergency escalations through ordinary phone calls. This settles who maintains the maps.
Satisfied by	§3.1.3.2 Stakeholders, WA7

2.1.2 DI2: Connected buildings

Original statement	"…going from one location to another in the same or different buildings that are connected to each other."
Classification	Inconsistent

Issue	The domain is described as “indoors” (a multi-floor building the user is familiar with), but this objective adds travel between buildings without saying whether the connectors (skybridges, tunnels, outdoor walkways) are in scope.
Option 1	A single building only.
Option 2	Multiple buildings, joined only by **indoor** connectors that are mapped.
Option 3	The whole campus, including outdoor paths between buildings.
Decision	Option 2
Rationale	This keeps the definition’s own objective (connected buildings) while staying indoors. Outdoor travel adds GPS, traffic, and weather, which are outside an indoor app.
Satisfied by	ID1, FR8

2.1.3 DI3: Destinations inside shared spaces

Original statement	Masterlist: “Navigation is limited for going from one room/location to another, NOT inside each room.”
Classification	Ambiguous
Issue	Some locations can only be reached by walking **through** another room (e.g., a study room inside a library, or an office inside a department suite). It isn’t clear whether routing through a room breaks the “not inside rooms” rule.
Option 1	Never route through any room; such destinations are unreachable.
Option 2	Route through rooms the map tags as **public pass-through** spaces (library, lobby, atrium); never through private rooms.
Decision	Option 2
Rationale	Option 1 would leave real destinations unreachable. Public pass-through spaces act like hallways, so the room-to-room rule still holds for private rooms.
Satisfied by	ID2, FR3

2.1.4 DI4: Navigating between floors

Original statement	“…the domain will be indoors, which can consist of multiple floors…”
Classification	Incomplete
Issue	The domain mentions multiple floors, but no requirement covers changing floors: finding the elevator or stairs, choosing between them, or confirming the user reached the right floor. The Masterlist assumes users can **operate** an elevator or stairs, but not that they can **find** one.
Option 1	Leave floor changes entirely to the user.
Option 2	THEIA guides the user to the elevator or stairs (by preference), says which floor to go to, and confirms the floor afterwards from beacons. The user operates the elevator or stairs.
Decision	Option 2
Rationale	Finding the elevator is the hard part for a blind user, not operating it (Scenario 3: Mark). Beacons confirm the floor on entry and exit, so the user never walks a wrong floor without knowing it.
Satisfied by	ID3, FR8

2.1.5 DI5: “Use as many sensors as possible”

Original statement	“…use of as many sensors that are available on an advanced smartphone would make your app more helpful.”
Classification	Ambiguous, Unsound

Issue	"As many as possible" isn't a testable requirement. Running every sensor all the time drains the battery and raises privacy concerns (camera, microphone), and some sensors add nothing for indoor navigation.
Option 1	Use every available sensor all the time.
Option 2	Use a defined set of sensors, each tied to a function.
Decision	Option 2
Rationale	Each sensor must justify itself: Bluetooth (beacons → location), accelerometer and gyroscope (steps, falls), compass (heading), microphone (voice commands), speaker and vibration (output). The camera is used only when the user chooses to send a photo in an emergency.
Satisfied by	WA9, NFR5

2.2 Issues with "Functional Requirements"

2.2.1 FRI1: Suggesting the next action from schedule or habit

Original statement	"Figuring out what the next action(s) would be, based on the user's schedule or habit, and suggesting/accepting the user's choice." Also bullet 1: "…suggest or confirm a possible destination location, utilizing the user's routine schedule or habit."
Classification	Ambiguous, Redundant
Issue	It doesn't say how THEIA "figures out" the next action: from location history, a schedule, or caretaker setup. The same idea also appears twice (bullets 1 and 7), which is redundant.
Option 1	Learn habits automatically from location history.
Option 2	Use a schedule and favorites entered by the user or caretaker.
Option 3	Both.
Decision	Option 2 (merge the two bullets into one requirement)
Rationale	An entered schedule is predictable and avoids tracking the user's movements. The user always confirms, so a wrong suggestion costs only one "No." Automatic learning can be reconsidered in Phase II.
Satisfied by	FR13

2.2.2 FRI2: Emergency calls when location is lost

Original statement	"Placing emergency calls and messages, possibly after detecting a fall or when the system has lost its current location."
Classification	Unsound, Incomplete
Issue	Losing the location happens routinely (between beacons, phone in a pocket). Treating that as an emergency would cause frequent false alarms. The statement also doesn't say who is contacted, or whether the user confirms first.
Option 1	Automatically call for help whenever location is lost or a fall is detected.
Option 2	Treat lost location as **off-route** and try to recover first. Contact someone only when the user says "Help," or after a fall with no response within 30 s. Text the caretaker first and offer campus police.
Decision	Option 2
Rationale	A 30-second check after a fall, and a cancel window after "Help," prevent false alarms. The outgoing text carries the last confirmed beacon location, so the helper knows where to go instead of searching a whole floor (Scenario 2).
Satisfied by	FR9, FR10

2.2.3 FRI3: Distance and direction instructions

Original statement	"Telling the user to walk a distance (e.g., 2 minutes before turning, or walk for 30 steps, etc.)" and "Telling the user to stop at the right place to turn."
Classification	Ambiguous, Incomplete
Issue	It isn't clear which unit to use (minutes, steps, meters) or how early to warn before a turn. It also doesn't cover non-straight paths, such as a curved hallway or walking around a pillar or fountain.
Option 1	Distances in meters or feet.
Option 2	Steps plus approximate seconds, with left/right/slight-left directions.
Option 3	Clock-face directions ("go to 2 o'clock").
Decision	Option 2, plus a warning about 5 steps before each turn, "stop" at the turn, and descriptions of curves and fixtures to go around
Rationale	Steps are relative to the body and many blind users already count them, while meters are hard to judge without sight. Step length is calibrated per user. The early warning gives time to slow down, fixing Stevie's early turn (Scenario 1).
Satisfied by	FR4, FR5

2.2.4 FRI4: Detecting obstacles

Original statement	"Detecting obstacles and telling the user what to do in order to avoid collision."
Classification	Unsound, Ambiguous
Issue	A phone in a pocket or hand can't reliably detect obstacles with its sensors, and the statement doesn't say which obstacles count. Promising detection the app can't deliver is a safety risk.
Option 1	Camera-based obstacle detection (the user holds the phone up).
Option 2	Static obstacles stored in the building map, plus blockages reported by users and staff. Moving obstacles are left to the cane or guide dog.
Decision	Option 2
Rationale	The cane and guide dog already handle moving and nearby obstacles. THEIA is responsible only for static obstacles in the map and reported blockages. This keeps the phone out of the user's hands, which are already busy with the cane or the dog's harness.
Satisfied by	FR6, FR7, WA5

2.2.5 FRI5: Presenting route options

Original statement	"Figuring out the routes to reach the destination, and informing the user of the options (if there are more than one), and accepting user's preference."
Classification	Incomplete
Issue	It doesn't say how many options to give or how to describe them by voice, since the user can't see a map. A long list read aloud is hard to remember.
Option 1	Always pick one route automatically.
Option 2	Read up to 3 routes, each with estimated time, number of turns, and floor changes; the user picks by number.
Decision	Option 2
Rationale	Three options can be held in memory. The summary (time, turns, elevator or stairs) is what users need to choose. If the user doesn't answer within 10 s, THEIA picks the best route for the current mode.
Satisfied by	FR3

2.2.6 FRI6: Missing “Where am I?” function

Original statement	“More that you can think of.”
Classification	Incomplete
Issue	The list has no way for the user to simply ask their current location, which is the first thing a blind user needs when disoriented. It’s also needed before starting a route and when reporting a location to a helper.
Option 1	Announce location only during navigation.
Option 2	A “Where am I?” voice command (and gesture) available at any time.
Decision	Option 2
Rationale	It’s cheap to provide (beacons already give the location) and supports recovery when lost (Scenario 2).
Satisfied by	FR1

2.3 Issues with “Non-functional Requirements”

2.3.1 NFRI1: “Ubiquitous”

Original statement	“The system shall be ubiquitous.”
Classification	Ambiguous
Issue	“Ubiquitous” isn’t defined or testable. It could mean every building, every phone, with or without internet, or always running.
Option 1	Works everywhere (not feasible without beacons and maps).
Option 2	Works in every **mapped** building, including without internet, and locates the user quickly.
Option 3	Works on every phone model.
Decision	Option 2
Rationale	It’s measurable: maps are stored on the phone, location updates within 2 s of entering a beacon’s range, and THEIA can be launched from the lock screen by voice. Emergency texts still need cellular.
Satisfied by	NFR5

2.3.2 NFRI2: “Fastest route”

Original statement	“The system shall lead the user through the fastest route.”
Classification	Ambiguous
Issue	“Fastest” isn’t defined. It could mean shortest distance, fewest turns, or least time. For a blind user, a short route with many turns or an elevator wait may not be the quickest.
Option 1	Shortest distance.
Option 2	Fewest turns.
Option 3	Least estimated walking time over accessible segments, adding time for turns and elevator waits.
Decision	Option 3
Rationale	Time is what the user actually cares about when late for class. Adding time for each turn reflects how blind users slow down at turns. Scenario 3 asks for the shortest accessible route with minimal turns; counting turn and elevator time covers both.
Satisfied by	NFR2

2.3.3 NFRI3: Safe, fast, and comfortable navigation

Original statement	“…safe navigation, fast navigation, and comfortable navigation” (§1), and the matching NFRs in §3.

Classification	Incomplete, Inconsistent (they can conflict)
Issue	None of the three is defined in terms of what the app does, and they can conflict: the fastest route may use stairs or a crowded atrium, which is less safe or comfortable. Nothing says which one wins.
Option 1	One fixed balance of the three, decided by the app.
Option 2	The user picks a mode (Fast, Safe, or Comfortable), and the basic safety rules apply in every mode.
Decision	Option 2
Rationale	Users' needs differ, and even one user's needs change between trips (late for class vs. tired). Keeping safety rules in every mode means choosing "Fast" never makes a route unsafe.
Satisfied by	FR12, NFR1, NFR3

2.3.4 NFRI4: Who customizes the app

Original statement	"The system shall be customizable to every user: e.g. volume, the interval of instructions, etc." vs. §1: "a caretaker … who sets the configuration of the app."
Classification	Inconsistent
Issue	§1 says the caretaker sets the configuration, while §3 says it is customizable to every user. It's unclear who may change which settings.
Option 1	Caretaker only.
Option 2	User only.
Option 3	Two levels: the caretaker sets safety settings (emergency contacts, fall detection, default mode) behind a PIN, and the user changes comfort settings (volume, speech rate, interval, current mode) by voice at any time.
Decision	Option 3
Rationale	Settings are split by their effect. Comfort settings (volume, speech rate) change often and are harmless to change, so the user controls them by voice. Safety settings (emergency contacts, fall detection) sit behind a caretaker PIN so they can't be turned off by accident (Scenario 4).
Satisfied by	FR11, NFR6

2.3.5 NFRI5: Alerting nearby pedestrians

Original statement	"The system shall help the user safely navigate indoors."
Classification	Incomplete
Issue	Safety could include making nearby pedestrians aware of the user so they step aside, but the definition doesn't mention people around the user.
Option 1	THEIA plays an audible alert to warn nearby pedestrians.
Option 2	Treat moving people as out of scope; the cane or guide dog already signals the user to others.
Decision	Option 2
Rationale	The Masterlist puts moving obstacles (pedestrians) out of scope. An alert sound would be socially awkward for the user and could compete with THEIA's own voice instructions. It can be revisited in Phase II.
Satisfied by	WA5 (documented as out of scope), FR6

2.3.6 NFRI6: "Usable for blind people"

Original statement	"The system shall be usable for blind people."
Classification	Ambiguous

Issue	“Usable” isn’t defined or measurable. The definition mentions voice recognition, but voice fails in noisy hallways, and nothing says how the user interacts without the screen.
Option 1	Voice only.
Option 2	Voice plus a standard screen reader over a normal visual interface.
Option 3	Every function is available by voice **and** by a simple gesture, with audio and vibration feedback, and nothing requires reading the screen.
Decision	Option 3
Rationale	Gestures back up voice in noise, vibration backs up audio when it’s hard to hear, and the result is testable: core tasks can be completed without sighted help, and navigation starts within 2 voice commands.
Satisfied by	NFR4

2.3.7 NFRI7: “Easily extensible”

Original statement	“The system shall be easily extensible to accommodate the following typical variations: variations in interface, language, definitive needs of the user, new features, new sensors and hardware, etc.”
Classification	Ambiguous
Issue	“Easily extensible” has no measure, and the list ends in “etc.” Without limits, this could mean anything.
Option 1	Every variation needs a new app release.
Option 2	Define extension points: new buildings via map data, new languages via resource files, new sensors via a sensor module.
Decision	Option 2
Rationale	Adding a building is the most common change and shouldn’t need an app update. Separating languages and sensors makes the other listed variations testable.
Satisfied by	NFR7

2.4 Summary of Issues

Section	Issues	Count
2.1 Domain / Stakeholders / Objectives	DI1–DI5	5
2.2 Functional	FRI1–FRI6	6
2.3 Non-Functional	NFRI1–NFRI7	7
Total		**18**

3. WRS

3.1 W (World)

3.1.1 Problems

Problem ID	Problem Description	Corresponding Goals
P1	A blind user inside a building can’t easily tell where they are or which direction leads to the destination. A cane or guide dog gives no room-level information.	G1, G2
P2	A blind user doesn’t know how far to walk, or exactly when to turn or stop. This leads to wrong turns and collisions (S1: Stevie turns too early and hits the corridor corner).	G3

Problem ID	Problem Description	Corresponding Goals
P3	Static obstacles and temporary blockages (construction, closed hallways) on a route aren't known in advance. A cane only detects them at arm's length.	G4
P4	Changing floors or moving into a connected building is hard: finding the elevator or stairs, and knowing which floor you're on (S3).	G5
P5	When a blind user gets lost or hurt, they can't tell helpers where they are, so help arrives slowly (S2: "somewhere on the 3rd floor").	G6
P6	Users have different needs (speed vs. safety vs. comfort, volume, instruction frequency), and existing aids are hard to adjust without sighted help (S4).	G7
P7	Typical smartphone apps depend on the screen. Blind users can't read it, and voice input can fail in noisy places.	G8
P8	Users repeat the same daily trips (classes, office, cafeteria) and have to start each one from scratch.	G9

3.1.2 Goals

Goal ID	Goal Description	Backward Traceability	Forward Traceability
G1	The user can find out their current location (building, floor, nearest room) at any time.	P1	IFO1, INFO5
G2	The user reaches a chosen destination by a route they accept.	P1	IFO2, INFO2
G3	The user gets timely and precise walk, turn, and stop instructions.	P2	IFO3, INFO1
G4	The user avoids known obstacles and blocked paths.	P3	IFO4, INFO1
G5	The user can change floors and move between connected buildings.	P4	IFO5
G6	An off-route user is guided back, and when help is really needed it reaches them quickly with their location.	P5	IFO6, INFO1
G7	The app adapts to each user, and it is clear what the user can change and what the caretaker controls.	P6	IFO7, INFO3, INFO6, INFO7
G8	Every function can be used without seeing the screen.	P7	INFO4
G9	Frequent trips can be started quickly.	P8	IFO8

3.1.3 Improved Understanding of Domain, Stakeholders, Functional and Non-Functional Objectives

3.1.3.1 Improved Domain

ID	Improved Domain Description	Resolves
ID1	THEIA is used **indoors** in university campus buildings that have been mapped and fitted with beacons. This includes buildings joined by **indoor connectors** (skybridge, tunnel, shared hallway), as long as the connector is mapped. Outdoor walking between unconnected buildings is out of scope.	DI2

ID	Improved Domain Description	Resolves
ID2	Navigation goes from one location (room, office, restroom, elevator, entrance) to another. It does **not** cover locations inside a room (chairs, desks). A route may pass **through** a room only if the map tags it as a public pass-through space (e.g., a library or lobby). Private rooms are never used as pass-throughs.	DI3
ID3	Buildings may have multiple floors. THEIA guides the user **to** the elevator or stairs and confirms the new floor afterwards. The user operates the elevator or stairs on their own.	DI4

World assumptions. Based on the World Assumptions Masterlist v1.0, plus our additions marked (+):

ID	Assumption
WA1	Primary users can hear, understand, and speak English, are reasonably familiar with the building, and already use smartphone accessibility features (voice commands, volume and home buttons).
WA2	Primary users can operate a normal elevator and/or use stairs. (+) Some users may set a “no stairs” preference.
WA3	Users may use a cane or guide dog alongside THEIA. THEIA complements these aids and doesn’t replace them.
WA4	Every room and significant location has a wireless beacon that broadcasts its ID (e.g., “Room 301”). (+) Corridor junctions, elevator lobbies, and stair entrances also have beacons.
WA5	Only **static** obstacles are in scope (pillars, trash cans, benches, corners, construction barriers). Moving obstacles such as pedestrians are out of scope, and the cane or dog handles them. (Resolves NFRI5)
WA6	Emergency dial-out is necessary. The phone has cellular service for calls and SMS.
WA7	(+) The campus accessibility office keeps a digital map of each building up to date: floor plans, beacon positions, walkable segments, static obstacles, accessible/inaccessible segments, busy or noisy areas, and temporary closures.
WA8	Caretakers are familiar with regular smartphone apps and can hear, understand, and speak English.
WA9	(+) Phone sensors used are Bluetooth (beacons), accelerometer and gyroscope (steps, falls), compass (heading), microphone (voice), speaker/earphone, and vibration motor. The camera is used only if the user chooses to send a photo in an emergency. (Resolves DI5)

3.1.3.2 Stakeholders

Resolves DI1.

Stakeholder	Type	Role / Responsibilities in THEIA
Blind or visually impaired user (student, staff, visitor)	Primary	Chooses destinations, follows guidance, adjusts personal preferences by voice, asks for help.
Caretaker (family member or assistant)	Secondary	Sets up the app and the safety settings (emergency contacts, fall detection, default mode). Receives off-route and emergency texts and comes to help when needed.
Accessibility services staff	Secondary	Create and maintain building maps and beacon lists, and mark temporary closures and obstacles. They also receive the blockage reports users send.
Campus police	Secondary	Receive emergency calls placed through the phone’s normal calling. No special system integration; THEIA reads or texts the location so the user can tell them.
Facilities / building management	Secondary	Install and maintain beacons, and tell accessibility services about construction.
Development / maintenance team	Other	Add new buildings, languages, and sensor support (see NFR7).

3.1.3.3 Improved Functional Objectives

Objective ID	Objective Description	Alleviates Problems	Achieves Goals	Resolves Issues
IFO1	Determine the user's current location from beacons and announce it on request and on arrival.	P1	G1	FRI6
IFO2	Accept a destination (and optionally a starting point) by voice, compute routes, present the options by voice, and accept the user's choice.	P1	G2	FRI5
IFO3	Give step-by-step guidance with distances in steps and approximate time, warn before each turn, and say "stop" at the turn or destination.	P2	G3	FRI3
IFO4	Warn about known static obstacles and blocked segments on the route, and reroute around reported blockages.	P3	G4	FRI4, NFRI5
IFO5	Guide the user across floors (by elevator or stairs, per preference) and through mapped connectors between buildings, confirming the floor.	P4	G5	DI2, DI4
IFO6	Detect when the user is off route, try to recover, and contact the caretaker or campus police only when the user asks for help or a fall goes unanswered, with a chance to cancel.	P5	G6	FRI2
IFO7	Provide two-level settings (caretaker safety settings, user comfort settings) and three navigation modes (Fast, Safe, Comfortable).	P6	G7	NFRI3, NFRI4
IFO8	Suggest the next destination from the user's entered schedule and favorites. The user confirms or declines.	P8	G9	FRI1

3.1.3.4 Improved Non-Functional Objectives

Objective ID	Objective Description	Alleviates Problems	Achieves Goals	Resolves Issues
INFO1	**Safety:** routes and instructions minimize the risk of collision or falls. When safety and speed conflict, safety wins.	P2, P3, P5	G3, G4, G6	NFRI3
INFO2	**Fast navigation:** "fastest" means the least estimated walking time over accessible segments.	P1, P6	G2	NFRI2
INFO3	**Comfortable navigation:** prefer simpler, familiar, quieter routes when the user asks for it.	P6	G7	NFRI3
INFO4	**Usability for blind users:** every function is available by voice and simple gestures, with audio and haptic feedback. Nothing is visual-only.	P7	G8	NFRI6
INFO5	**Ubiquity:** THEIA works in every mapped building, including without an internet connection, and locates the user quickly.	P1	G1	NFRI1
INFO6	**Customizability:** output and pacing can be adjusted per user, and the change takes effect immediately.	P6	G7	NFRI4

Objective ID	Objective Description	Alleviates Problems	Achieves Goals	Resolves Issues
INFO7	**Extensibility:** new buildings, languages, and sensors can be added without redesigning the app.	P6	G7	NFRI7

3.2 RS (Requirements and Specifications)

3.2.1 Functional Requirements

FR1: Where am I

Description	When the user says “Where am I?” (or double-taps with two fingers), THEIA announces the building, floor, nearest room or landmark, and the direction the user is facing.
Specifications	FS1.1 Location comes from the strongest beacon signal received in the last 3 s. FS1.2 Facing direction comes from the compass (“You are facing the elevators”). FS1.3 If no beacon is received for 10 s, THEIA says “Location unknown. Last known location was ___, heard ___ ago” (see FR9). FS1.4 THEIA answers within 2 s.
Satisfies Issue	FRI6
Satisfies Objectives	IFO1
Scenario / Prototype	S2 / Screen 2 Navigation
Priority	High

FR2: Destination and starting point input

Description	THEIA asks “Where would you like to go?” The user gives a destination by voice (“Take me to Spark 233,” “I’m heading to lunch at the cafeteria”) or picks a saved favorite. The starting point is the user’s current location (FR1); the user may instead dictate or type a different starting point. THEIA repeats the destination and asks for confirmation.
Specifications	FS2.1 Confirmation is “Yes” or a double-tap, with one short vibration. FS2.2 If the name matches several places, THEIA lists up to 3 (“Did you mean Spark 233 or Spark 223?”). FS2.3 If the destination isn’t in the map, THEIA says so and asks again. FS2.4 If the user gives no starting point, the current beacon location is used; a dictated or typed starting point is checked against the map the same way as a destination.
Satisfies Issue	— (preliminary FR bullet 1)
Satisfies Objectives	IFO2
Scenario / Prototype	S1, S3, S4 / Screen 1 Voice Home
Priority	High

FR3: Route options

Description	THEIA computes up to 3 routes and reads each one’s estimated time, number of turns, and floor changes (elevator or stairs). The user picks one by saying its number.
Specifications	FS3.1 Routes use only walkable segments in the map, and public pass-through rooms only (ID2). FS3.2 Routes with stairs are excluded if “no stairs” is set. FS3.3 If the user gives no answer within 10 s, THEIA picks the best route for the current mode (FR12). FS3.4 If only one route exists, THEIA says so and starts. FS3.5 If the user names the elevator or stairs (“via elevator”), only routes using it are offered, and the first one offered is the shortest accessible route with the fewest turns. FS3.6 If no route exists (for example, every path is closed), THEIA says so and offers a route to the nearest reachable point.
Satisfies Issue	FRI5, DI3
Satisfies Objectives	IFO2
Scenario / Prototype	S3 / Screen 1 Voice Home

Priority	High

FR4: Step-by-step guidance

Description	THEIA gives each leg as “Walk about 20 steps, around 15 seconds, then turn left,” and describes the path shape, such as curving hallways or going around a pillar or fountain.
Specifications	FS4.1 Step length is calibrated per user during setup (default 0.7 m). FS4.2 A pre-turn alert is given 5 steps before the turn (“In 5 steps, turn left”). FS4.3 “Stop. Turn left now” is given at the turn, with one short vibration. FS4.4 Directions use left/right/slight left/slight right/straight. FS4.5 The user can say “Repeat” or double-tap at any time.
Satisfies Issue	FRI3
Satisfies Objectives	IFO3
Scenario / Prototype	S1, S3 / Screen 2 Navigation
Priority	High

FR5: Arrival

Description	THEIA announces arrival and the side of the door (“Spark 233 is on your right”), with one long vibration.
Specifications	FS5.1 Triggered when the destination beacon is the strongest one and the user is within about 2 steps of the door position in the map.
Satisfies Issue	FRI3
Satisfies Objectives	IFO3, IFO1
Scenario / Prototype	S1 / Screen 2 Navigation
Priority	High

FR6: Static obstacle warning

Description	THEIA warns about static obstacles recorded in the map ahead on the route and says how to avoid them (“Bench on your right in 5 steps, keep left”).
Specifications	FS6.1 The warning comes 5 steps before the obstacle. FS6.2 Only map-recorded static obstacles are announced. Moving obstacles are left to the cane or dog (WA5).
Satisfies Issue	FRI4, NFRI5
Satisfies Objectives	IFO4
Scenario / Prototype	S1 / Screen 2 Navigation
Priority	High

FR7: Blockage report and reroute

Description	When the user says “This way is blocked,” THEIA answers “OK. Finding a detour,” marks that segment closed for the current trip, computes a detour, and sends a blockage report to accessibility services.
Specifications	FS7.1 The detour starts within 3 s. FS7.2 Closures that staff enter in the map are avoided automatically on every route. FS7.3 If no safe detour exists, THEIA says so and offers Help (FR10).
Satisfies Issue	FRI4
Satisfies Objectives	IFO4
Scenario / Prototype	S2 / Screen 3 Off-route
Priority	Medium

FR8: Floor and building change

Description	THEIA guides the user to the elevator or stairs, by preference or choice ("THEIA, lead me to TFLO 131 via elevator"), says which floor to go to, and confirms the floor afterwards.
Specifications	FS8.1 Audio cues lead the user to the elevator doors; one short vibration confirms arrival at the elevator, and THEIA says "Call button on your right. Go to floor 1." FS8.2 After leaving the elevator or stairs, THEIA checks the floor from beacons, says "You are on floor 1," switches to that floor's map automatically, and continues guidance. If it's the wrong floor, it says so and guides the user back to the elevator or stairs. FS8.3 Mapped connectors between buildings are treated like hallways, and THEIA announces the crossing at both ends ("Entering the skybridge to ___", then "Entering building ___").
Satisfies Issue	DI2, DI4
Satisfies Objectives	IFO5
Scenario / Prototype	S3 / Screen 2 Navigation (elevator)
Priority	High

FR9: Off-route detection and recovery

Description	If the user leaves the planned route, THEIA gives three short vibrations, says "You may be off route. Say 1 to find a new route, or 2 to go back to ___," and either (1) reroutes from the current position or (2) guides the user back to the last known location.
Specifications	FS9.1 Off route means an unexpected beacon is received, or the next expected beacon isn't received within the expected time plus 30 s. FS9.2 The user answers by saying "1" or "2," or by tapping the top or bottom half of the screen. FS9.3 If there is no beacon at all, THEIA tells the user the last confirmed location and guides them along the nearest wall until a beacon is found. FS9.4 Losing location does **not** trigger an emergency call by itself. If recovery fails after 2 minutes, THEIA *asks* "Do you want me to text your caretaker your last location?"
Satisfies Issue	FRI2
Satisfies Objectives	IFO6
Scenario / Prototype	S2 / Screen 3 Off-route
Priority	High

FR10: Help and emergency

Description	When the user says "Help" (or holds two fingers on the screen for 2 s), the Help screen opens with a strong vibration and THEIA asks "Do you need help? Text caretaker, call caretaker, call campus police, or cancel?" and acts on the answer. If there is no answer, THEIA texts the caretaker the user's location automatically. THEIA also checks for falls.
Specifications	FS10.1 Every help text includes the building, floor, nearest room (last confirmed beacon), last-update time, and a map link. FS10.2 If the user gives no answer to the Help question within 10 s, THEIA texts the caretaker automatically. FS10.3 On a detected fall (accelerometer), THEIA gives a strong continuous vibration and says "Are you OK? I will text ___ in 30 seconds. Say Cancel to stop." If there's no response within 30 s, it texts the caretaker and then calls the first emergency contact. FS10.4 The user can say "Cancel" (or tap Cancel) to stop either countdown. FS10.5 If the caretaker doesn't answer the call, or doesn't reply to the text within 2 minutes, THEIA immediately offers to call campus police. FS10.6 Optionally, the user can send a photo from the phone camera. FS10.7 If there is no cellular signal, THEIA tells the user, retries every 30 s, and sends the text or places the call as soon as the phone connects.
Satisfies Issue	FRI2
Satisfies Objectives	IFO6
Scenario / Prototype	S2 / Screen 4 Help, Screen 4 Help (fall check)

Priority	High

FR11: Two-level settings

Description	The caretaker sets the **safety settings** (emergency contacts, fall detection on/off, default mode, “no stairs”). The user changes **comfort settings** by voice at any time (“THEIA, louder,” “Speak slower,” “Give instructions every 10 seconds”).
Specifications	FS11.1 Safety settings are changed from a caretaker screen protected by a PIN. FS11.2 The user cannot remove emergency contacts or turn off fall detection without the caretaker PIN. FS11.3 Every voice change is confirmed aloud (“Volume 80 percent”).
Satisfies Issue	NFRI4
Satisfies Objectives	IFO7
Scenario / Prototype	S4 / Screen 5 Settings
Priority	Medium

FR12: Navigation modes

Description	The user selects **Fast**, **Safe**, or **Comfortable** mode by voice before or during a trip. The caretaker’s default mode (FR11) is used until the user changes it.
Specifications	FS12.1 Fast: minimum estimated time (NFR2). FS12.2 Safe: avoids stairs and segments with recorded obstacles, and confirms every turn. FS12.3 Comfortable: fewest turns, prefers previously used routes and avoids areas the map tags as busy or noisy (e.g., the main lobby), at the cost of up to 25% more time. FS12.4 The safety rules in NFR1 apply in every mode.
Satisfies Issue	NFRI2, NFRI3
Satisfies Objectives	IFO7
Scenario / Prototype	S3, S4 / Screen 5 Settings, Screen 1 Voice Home
Priority	Medium

FR13: Schedule-based suggestion

Description	Using a class schedule and favorites entered by the user or caretaker, THEIA suggests the next destination about 10 minutes before a scheduled event (“Your next class is TFLO 131 at 10:10. Start navigation?”).
Specifications	FS13.1 The user answers “Yes” or “No”. Nothing starts without confirmation. FS13.2 In Phase I, THEIA does not learn habits automatically from location history (privacy and predictability). This can be revisited in Phase II.
Satisfies Issue	FRI1
Satisfies Objectives	IFO8
Scenario / Prototype	— / Screen 1 Voice Home
Priority	Low

3.2.2 Non-Functional Requirements

NFR1: Safety

Description	Routes never use map segments marked inaccessible or closed. Obstacle and turn warnings are given 5 steps ahead, early enough to act on. When speed and safety conflict, the safer option is chosen.
Acceptance criterion	Test routes in all three modes never use closed or inaccessible segments, obstacle and turn warnings come 5 steps ahead, and Help (FR10) works in every mode.
Operationalized FRs	FR4, FR6, FR7, FR10, FR12
Satisfies Issue	NFRI3

Satisfies Objective	INFO1
Constrains	FR3, FR8, FR9
Scenario / Prototype	S1, S2 / Screen 2 Navigation, Screen 4 Help
Priority	High

NFR2: Fastest route (definition)

Description	The “fastest” route is the one with the least estimated walking time over accessible segments, using the user’s walking speed plus about 5 s per turn and about 60 s per elevator wait. Shortest distance alone is not used.
Acceptance criterion	On a test map, the route ranked first in Fast mode has the lowest estimated time (walking time at the user’s speed, plus 5 s per turn, plus 60 s per elevator wait).
Operationalized FRs	FR3, FR12
Satisfies Issue	NFRI2
Satisfies Objective	INFO2
Constrains	FR3
Scenario / Prototype	S3 / Screen 1 Voice Home
Priority	Medium

NFR3: Comfort

Description	In Comfortable mode, routes prefer fewer turns, routes the user has taken before, and areas not tagged busy or noisy, with at most 25% extra time.
Acceptance criterion	On a test map, Comfortable mode picks a route that takes at most 25% longer than the fastest route, and no other route within that limit has fewer turns or passes fewer busy or noisy areas.
Operationalized FRs	FR12
Satisfies Issue	NFRI3
Satisfies Objective	INFO3
Constrains	FR3
Scenario / Prototype	S4 / Screen 5 Settings
Priority	Medium

NFR4: Usability for blind users

Description	(a) Every function is available by voice **and** by a simple gesture, so it still works where voice recognition fails because of noise. (b) No function requires reading the screen or tapping a small target. (c) Every action gets both audio and haptic confirmation. (d) Navigation can be started with at most 2 voice commands. (e) Large high-contrast text is shown for low-vision users, but no information is shown only visually.
Acceptance criterion	Blind and visually impaired testers complete FR1, FR2, FR3, and FR10 without sighted help, and start navigation within 2 voice commands.
Operationalized FRs	All FRs
Satisfies Issue	NFRI6
Satisfies Objective	INFO4
Constrains	FR1–FR13
Scenario / Prototype	S1–S4 / All screens
Priority	High

NFR5: Ubiquity / availability

Description	THEIA works in every mapped building. Building maps are stored on the phone, so navigation works without internet (emergency calls and texts still need cellular). The location updates within 2 s of entering a beacon's range. THEIA can be launched from the lock screen with the phone's voice assistant ("Hey Siri / Hey Google, open THEIA").
Acceptance criterion	With airplane mode on (Bluetooth only), FR1–FR9 work in a mapped building, and the location updates within 2 s in 95% of trials.
Operationalized FRs	FR1, FR4, FR9
Satisfies Issue	NFRI1, DI5
Satisfies Objective	INFO5
Constrains	FR1–FR9
Scenario / Prototype	S1–S3 / Screen 2 Navigation
Priority	High

NFR6: Customizability

Description	The user can adjust volume, speech rate (0.5×–2×), instruction interval (every 5–30 s, or at turns only), haptic strength (off/low/high), and the current navigation mode. Changes take effect immediately. Safety settings, including the default mode, are caretaker-level (FR11).
Acceptance criterion	Each setting changes behavior immediately as specified and is kept after the app restarts.
Operationalized FRs	FR11, FR12
Satisfies Issue	NFRI4
Satisfies Objective	INFO6
Constrains	FR4, FR5, FR6
Scenario / Prototype	S4 / Screen 5 Settings
Priority	Medium

NFR7: Extensibility

Description	New buildings are added by loading map data, without an app update. New languages are added through text and voice resource files. New sensors (e.g., smart-cane input) plug into a sensor module without changing the navigation logic.
Acceptance criterion	Loading a test building's map makes it routable without a new app release.
Operationalized FRs	FR1, FR3, FR6
Satisfies Issue	NFRI7
Satisfies Objective	INFO7
Constrains	—
Scenario / Prototype	—
Priority	Low

4. Preliminary Prototype and User Manual

The prototype is a non-functional mock-up (*Theia Prototype Mock-Up*, submitted separately as .pptx and .pdf). The preliminary user manual (*Theia-Preliminary User Manual v1.1*) is also submitted separately. This section summarizes both and shows how they map to the requirements in §3.

4.1 Interaction Model

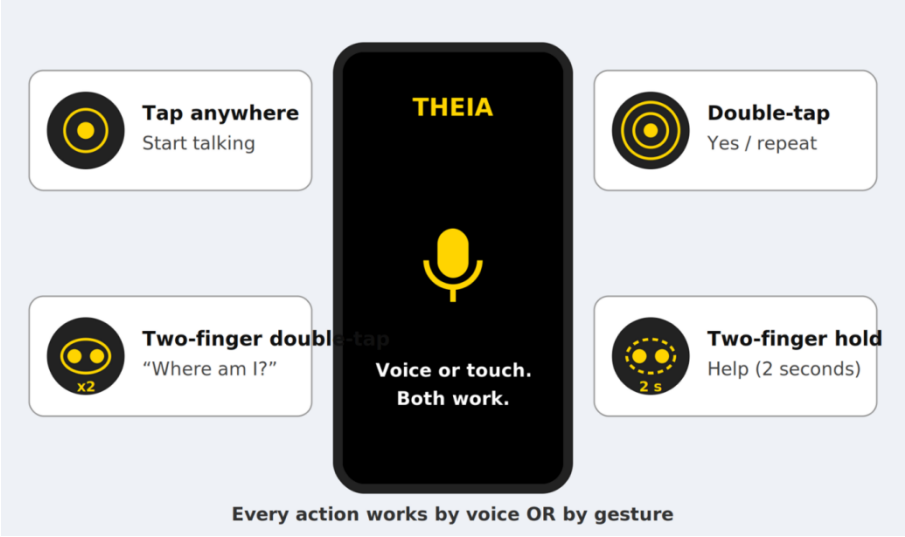
THEIA is used by voice, touch, and vibration; the screen is a bonus, not a must (NFR4). Every action can be done by voice or by one of four gestures, and every response is spoken and felt.

Gesture	Meaning	Requirement
Tap anywhere (Voice Home)	Start talking	FR2
Double-tap	Yes / repeat the last instruction	FR2, FR4
Two-finger double-tap	“Where am I?”	FR1
Two-finger hold (2 s)	Help	FR10
Tap top half / bottom half (Off-route screen)	Option 1 / option 2	FR9

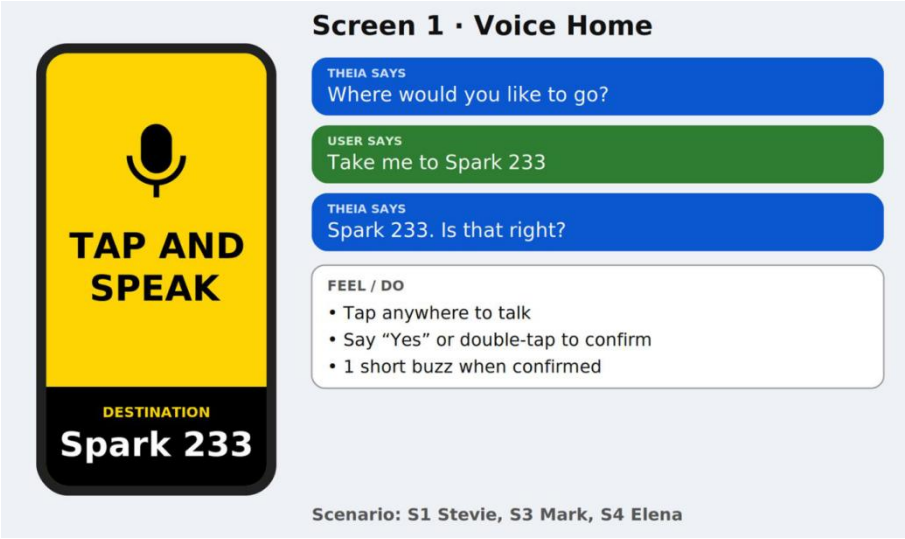
Vibration	Meaning	Requirement
1 short buzz	Turn now; confirmation; arrived at the elevator	FR2, FR4, FR8
1 long buzz	You have arrived	FR5
3 buzzes	Off-route warning	FR9
Strong continuous buzz	Help screen or fall check opened	FR10

4.2 Prototype Screens

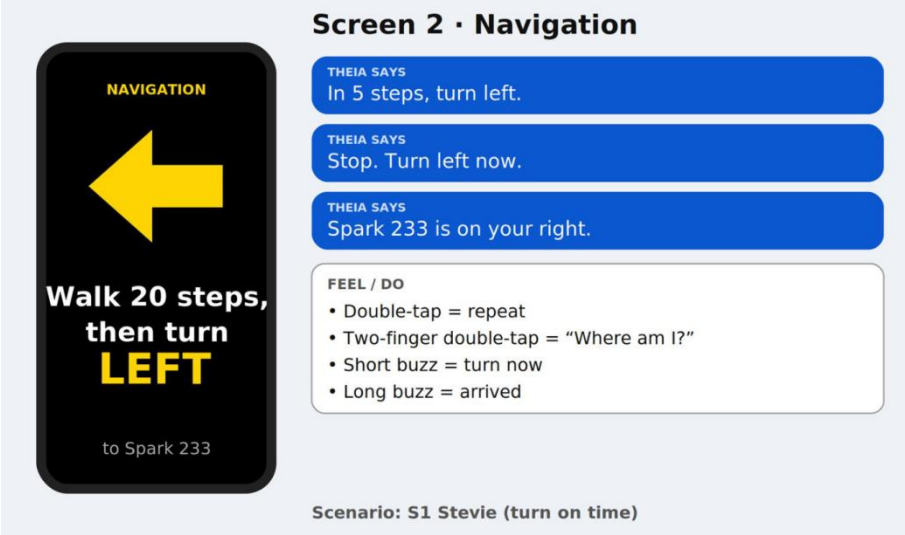
Screen	Purpose	Reached without sight by	Scenario	Requirements
1 Voice Home	Destination input, route choice, class reminder	Opening THEIA (voice assistant) or tapping anywhere	S1, S3, S4	FR2, FR3, FR12, FR13
2 Navigation	Walk, turn, stop, arrival, obstacle warnings	Confirming a destination	S1	FR1, FR4, FR5, FR6
2 Navigation (elevator)	Elevator guidance and floor check	Route reaches an elevator	S3	FR8
3 Off-route	Reroute or go back; blocked path	Automatic (3 buzzes) or “This way is blocked”	S2	FR7, FR9
4 Help	Text/call caretaker, call campus police, cancel	“Help” or two-finger hold	S2	FR10
4 Help (fall check)	30-second fall countdown	Automatic on a detected fall	S2	FR10
5 Settings	User comfort settings; caretaker PIN settings	“THEIA, louder” etc.; caretaker opens with PIN	S4	FR11, FR12, NFR6



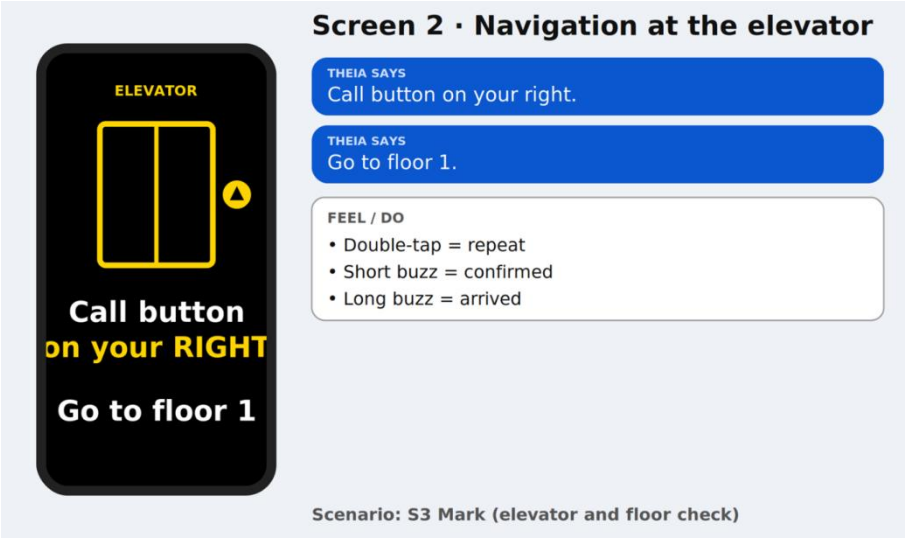
How the user controls THEIA: four gestures (tap anywhere, double-tap, two-finger double-tap, two-finger hold).



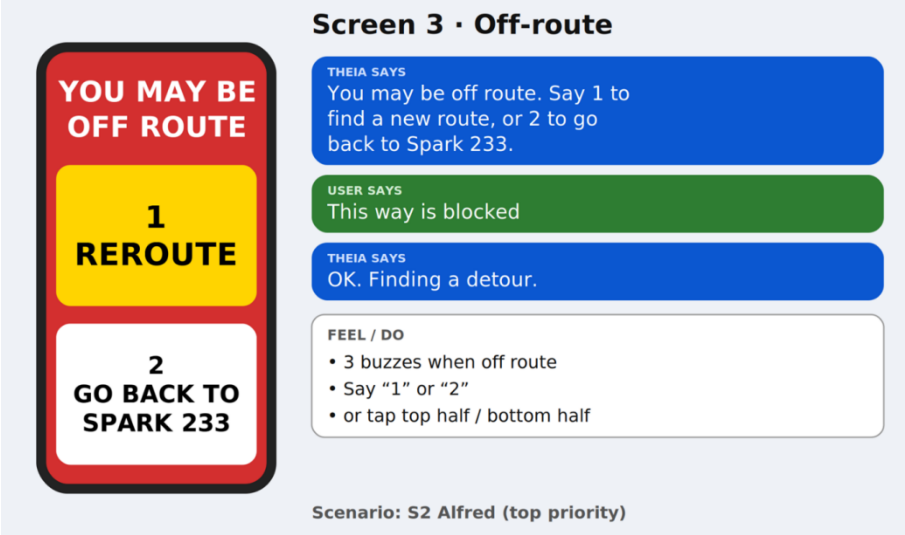
Screen 1, Voice Home: THEIA asks “Where would you like to go?”, the user says “Take me to Spark 233”, THEIA confirms.



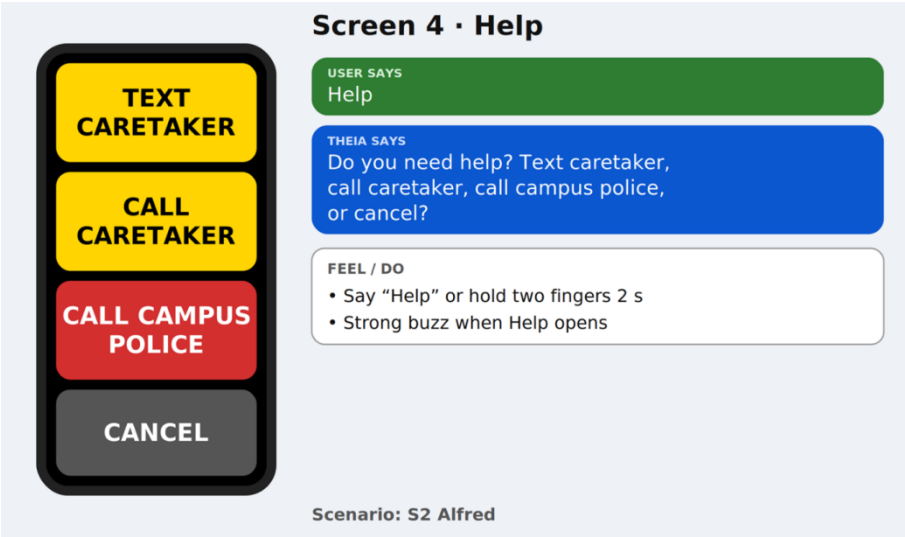
Screen 2, Navigation: "Walk 20 steps, then turn left", the 5-step turn warning, "Stop. Turn left now", and arrival.



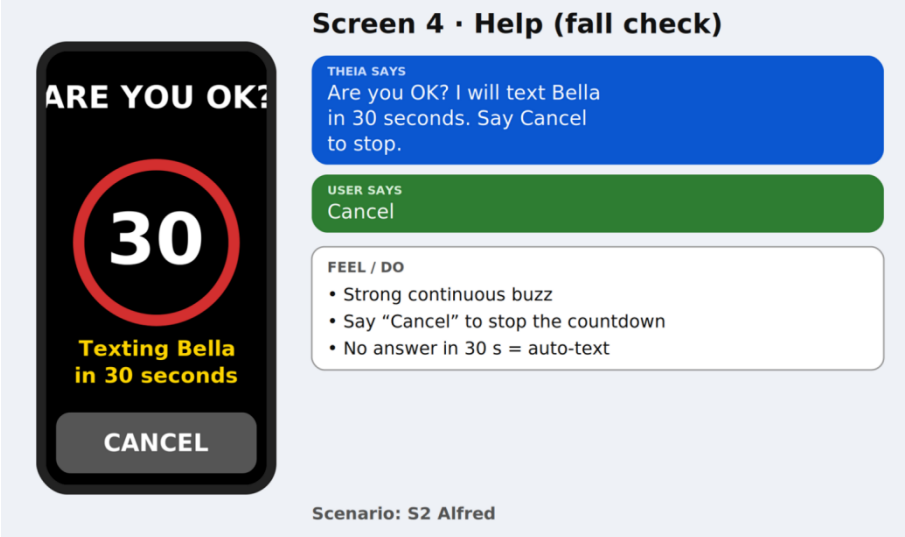
Screen 2, Navigation at the elevator: "Call button on your right. Go to floor 1."



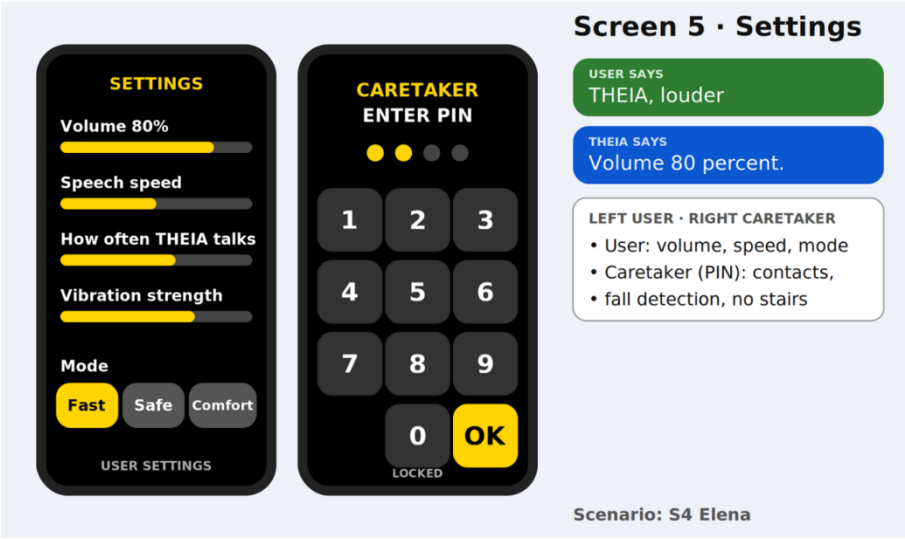
Screen 3, Off-route: "You may be off route. Say 1 to find a new route, or 2 to go back to Spark 233", and the "This way is blocked" command.



Screen 4, Help: text caretaker, call caretaker, call campus police, or cancel.



Screen 4, Help (fall check): "Are you OK? I will text Bella in 30 seconds. Say Cancel to stop."



Screen 5, Settings: user settings (volume, speech speed, instruction frequency, vibration strength, mode) next to the caretaker PIN keypad.

4.3 Preliminary User Manual

The user manual is written to be read aloud by a screen reader: numbered steps, no instructions that depend on seeing the screen, and every step says what the user will hear or feel. It is organized around the TO-BE scenarios.

Manual section	Scenario	Requirements
1 About THEIA	—	NFR4, NFR5
2 First-Time Setup (Caretaker)	S4	FR11, FR13, FS4.1
3 Basic Controls	All	FR1, FR2, FR4, NFR4
4 Going to a Room	S1 Stevie	FR2, FR3, FR4, FR5, FR6
5 Changing Floors by Elevator	S3 Mark	FR3, FR8
6 If You Are Lost or Need Help	S2 Alfred	FR7, FR9, FR10

Manual section	Scenario	Requirements
7 Settings and Navigation Modes	S4 Elena	FR11, FR12, FR13, NFR6
8 Quick Reference	All	FR1–FR12
9 Limitations	—	WA3, WA5, WA6, ID2

4.4 Voice Commands

User says	THEIA does	Requirement
“Take me to ___” / “I’m heading to ___”	Starts a route	FR2
“Start from ___”	Uses a different starting point	FR2
“Yes” / “No”	Confirms or declines	FR2, FR13
“1”, “2”, “3”	Picks a route or an off-route option	FR3, FR9
“Where am I?”	Tells building, floor, nearest room, and which way you face	FR1
“Repeat”	Says the last instruction again	FR4
“This way is blocked”	Finds a detour	FR7
“Help”	Opens the Help screen	FR10
“Cancel”	Stops the current action or countdown	FR10
“Louder” / “Quieter”, “Speak faster” / “Speak slower”, “Instructions every ___ seconds”	Changes a comfort setting	FR11, NFR6
“Fast mode” / “Safe mode” / “Comfortable mode”	Changes the route style	FR12

5. Traceability

5.1 Forward and Backward Traceability

Forward, read left to right: Problem → Goal → Objective → Requirement. Backward, read right to left: every requirement traces to an Objective, Goal, and Problem.

Problem	Goal	Functional Obj.	Non-Functional Obj.	FR	NFR	Issues (§2)	Scenario
P1	G1	IFO1	INFO5	FR1, FR5	NFR5	FRI6, NFRI1, DI5	S2
P1	G2	IFO2	INFO2	FR2, FR3	NFR2	FRI5, DI3, NFRI2	S1, S3, S4
P2	G3	IFO3	INFO1	FR4, FR5	NFR1	FRI3	S1
P3	G4	IFO4	INFO1	FR6, FR7	NFR1	FRI4, NFRI5	S1, S2
P4	G5	IFO5	—	FR8	NFR1	DI2, DI4	S3
P5	G6	IFO6	INFO1	FR9, FR10	NFR1	FRI2	S2
P6	G7	IFO7	INFO3, INFO6, INFO7	FR11, FR12	NFR3, NFR6, NFR7	NFRI3, NFRI4, NFRI7	S4
P7	G8	—	INFO4	all	NFR4	NFRI6	S1–S4
P8	G9	IFO8	—	FR13	—	FRI1	—
—	—	—	—	—	—	DI1 → resolved by §3.1.3.2 Stakeholders	S2

5.2 Issue → Resolution

Every §2 issue lands somewhere in §3.

Issue	Resolved in
DI1 Stakeholder roles	3.1.3.2, WA7
DI2 Connected buildings	ID1, FR8
DI3 Shared spaces	ID2, FR3
DI4 Multi-floor	ID3, FR8
DI5 Sensors	WA9, NFR5
FRI1 Next action / schedule	IFO8, FR13
FRI2 Emergency on lost location	IFO6, FR9, FR10
FRI3 Distance and direction	IFO3, FR4, FR5
FRI4 Obstacle detection	IFO4, FR6, FR7
FRI5 Route options	IFO2, FR3
FRI6 “Where am I”	IFO1, FR1
NFRI1 Ubiquitous	INFO5, NFR5
NFRI2 Fastest	INFO2, NFR2
NFRI3 Safe/Fast/Comfortable	INFO1, INFO3, FR12, NFR1, NFR3
NFRI4 Who customizes	IFO7, FR11, NFR6
NFRI5 Alerting pedestrians	WA5 (out of scope), FR6
NFRI6 Usability	INFO4, NFR4
NFRI7 Extensibility	INFO7, NFR7

5.3 Scenario → Requirements → Prototype → Manual

Scenario	TO-BE behavior	Requirements	Prototype screens	Manual
S1 Basic step-by-step guidance (Stevie)	Destination (and start point) by voice; “Walk ahead 10 steps, then turn left”; turn on time	FR2, FR3, FR4, FR5, FR6, NFR1	1, 2	§4
S2 Off-route recovery and emergency escalation (Alfred), **Top Priority**	“This way is blocked” → detour; missed beacon → “may be off route” → reroute or back to Spark 233; “Help” or unanswered fall → text to Bella with location; campus police offered	FR1, FR7, FR9, FR10, NFR1	3, 4, 4 (fall check)	§6
S3 Multi-floor navigation by elevator (Mark)	“Lead me to TFLO 131 via elevator”; audio to the elevator, buzz on arrival; floor confirmed and map switched	FR2, FR3, FR8, NFR2	1, 2 (elevator)	§5
S4 Personal settings and navigation modes (Elena)	“I’m heading to lunch at the cafeteria” + double-tap; “THEIA, louder”, “instructions every 10 seconds”; route avoids the busy lobby; caretaker contacts stay locked	FR2, FR11, FR12, NFR3, NFR6	1, 5	§7

5.4 Preliminary Definition → WRS Coverage

Every statement in the THEIA Preliminary Definition v1.0 (§1 domain, stakeholders, and objectives; §2 functional requirements; §3 non-functional requirements) and where this WRS covers it. Adapted from Justus Jones's §3 draft and remapped to this document's IDs.

Def. §	Statement (short)	Covered by
1	The domain is indoors: multiple floors with classrooms, offices, bathrooms, lounges, and elevators	§3.1.3.1, ID1, ID3
1	Primary stakeholder: a blind person navigating indoors	§3.1.3.2
1	The caretaker sets the configuration of the app	§3.1.3.2, FR11
1	The caretaker comes to help in an emergency (lost or injured)	FR9, FR10
1	Accessibility staff and police as secondary stakeholders	§3.1.3.2, WA7
1	Navigate between locations in the same or connected buildings	ID1, FR3, FR8
1	Non-functional objectives: safe, fast, and comfortable navigation	INFO1–INFO3, NFR1–NFR3
1	Usability is a top priority; the user can't read the screen; use voice recognition	INFO4, NFR4 (High)
1	Use as many smartphone sensors as possible	WA9, NFR5 (narrowed per DI5)
2	Accept the destination; suggest or confirm one from the schedule or habit	FR2, FR13
2	Find routes, present the options, accept the user's preference	FR3, FR12
2	Tell the user to walk a distance (minutes, steps)	FR4
2	Tell the user to stop at the right place to turn	FR4 (FS4.2, FS4.3)
2	Detect obstacles and say how to avoid them	FR6, FR7, WA5
2	Emergency calls and messages after a fall or a lost location	FR9, FR10 (lost location is handled as off-route first, per FRI2)
2	Suggest the next action from the schedule or habit	FR13 (merged with the first bullet, per FRI1)
2	"More that you can think of"	FR1, FR5, FR7, FR8, FR11
3	Help the user navigate safely	NFR1
3	Lead the user through the fastest route	NFR2
3	Lead the user through the most comfortable route	NFR3
3	Usable for blind people	NFR4
3	Ubiquitous	NFR5
3	Customizable to every user (volume, instruction interval)	FR11, NFR6
3	Easily extensible (interface, language, user needs, features, sensors, hardware)	NFR7 (limited to buildings, languages, and sensors, per NFRI7)

References

1. Cpt S 484 Team Project Specification, Phase I, v2.0 (course materials).
2. Cpt S 484 Project THEIA, Preliminary Definition, v1.0 (course materials).
3. CPTS 484 FAQ (course materials).
4. World Assumptions Masterlist, v1.0 (course materials).
5. WRS-template.rtf and WRS-Sample-Incomplete.docx (course materials).
6. Requirements Review Checklist (course materials).

7. Team EV, *Theia AS-IS and TO-BE v2.1 Final* (slides) and presentation video, https://youtu.be/0aHIRkDOK3E.
8. Team EV, *Theia Prototype Mock-Up* and *Theia-Preliminary User Manual v1.1*.
9. Team EV, *Project Plan, Final Phase I Version 1.2*.
10. Team EV, *AI Usage and Teamwork Report*.
11. Justus Jones, WRS Section 3 draft (WRS-Section3-Updated.md), Team EV repository, Oct 18, 2026.